

Dairy Farm Analysis Report

Introduction

This analysis examines the performance of a large-scale dairy operation across India with significant production capacity and market presence. The operation manages 238,000 cows across 15 locations, producing 10 different dairy products under 11 distinct brands. With total revenue of ₹58.73 million and a total value of ₹118.32 million, this enterprise demonstrates the substantial economic impact of organized dairy farming in the Indian agricultural sector.

Key Metrics

Cows: 238,000 cows across 15 locations.

Brands: 11 dairy brands.

Products: 10 different dairy products.

Total Revenue: ₹58.73 million

Total Value: ₹118.32 million

Detailed Analysis

Cows by Farm Size

- Large Farms: 81K (34.09%)
- Medium Farms: 79K (33.36%)
- Small Farms: 77K (32.55%)

Implications: The distribution among farm sizes is relatively balanced, with large farms slightly leading. This indicates that large farms may have more resources and access to technology, while small and medium farms hold significant contributions.

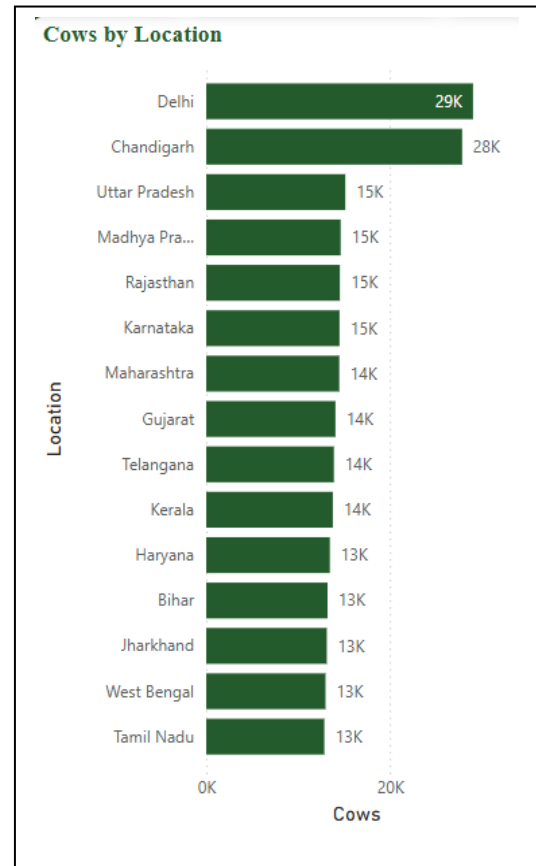
Recommendations:

- Provide funding or incentives for small and medium farms to adopt advanced farming methods.
- Share best practices from large farms with smaller ones to improve efficiency.
- Develop tailored programs to support farms based on their specific needs.

Cows by Location

The percentage distribution of the 238,000 cows across 11 locations as follows;

- Delhi: 12.18%
- Chandigarh: 11.76%
- Uttar Pradesh: 6.30%
- Madhya Pradesh: 6.30%
- Rajasthan: 6.30%
- Karnataka: 6.30%
- Maharashtra: 5.88%
- Gujarat: 5.88%
- Telangana: 5.88%
- Kerala: 5.88%
- Haryana: 5.46%
- Bihar: 5.46%
- Jharkhand: 5.46%
- West Bengal: 5.46%
- Tamil Nadu: 5.46%



Implications: Delhi and Chandigarh have the highest numbers of cows, indicating these regions may be focal points for dairy farming and production. However, other areas show potential for growth and expansion.

Recommendations:

- Focus on expanding dairy operations in regions with lower cow populations to balance supply and demand.
- Introduce tailored programs in Delhi and Chandigarh to maximize efficiency and production.
- Explore market opportunities in lower-performing regions such as Tamil Nadu and West Bengal.

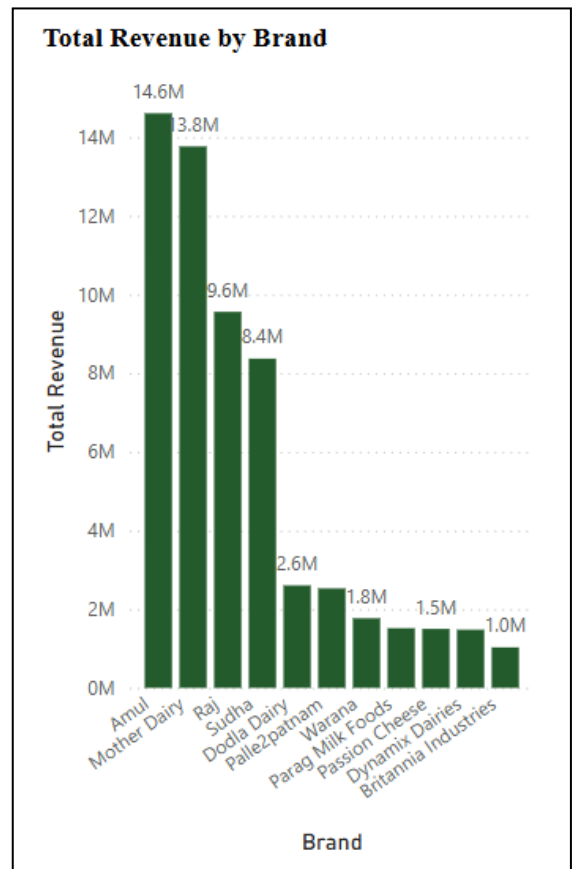
Total Revenue by Brand

Amul dominates with ₹14.6M (29.2% of total revenue). Mother Dairy generates ₹9.6M (19.2%), and Raj contributes ₹8.4M (16.8%) of the total revenue. Other brands like Ujala account for smaller shares, such as 5.44%.

Implications: High revenues by Amul and Mother Dairy suggest strong brand presence and customer trust.

Recommendations:

- Smaller brands should focus on marketing and improving quality to compete.
- Collaborative campaigns with established brands could boost visibility and revenue.



Storage Condition by Count

Refrigerated storage dominates with 56.9% of units. Frozen products comprise 23.93% of units. Ambient storage accounts for 9.29% of units. Polythene Packet and Tetra Pack make up 10.0% of units combined.

Implications:

- Heavy dependence on cold chain infrastructure
- Significant portion requires ambient storage, which is less costly

Recommendations:

- Invest in cold chain optimization to reduce costs
- Explore expanding ambient and tetra pack products to reduce dependency on cold chain.

Total Revenue by Location

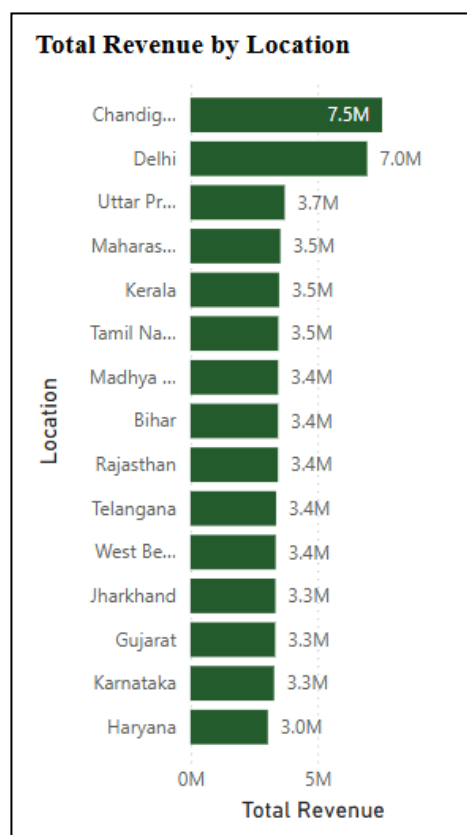
Chandigarh leads with ₹7.5M which amounts to 12.84% of total revenue.

Delhi follows closely with ₹7.0M (11.84%), and Uttar Pradesh contributes ₹3.7M (6.31%) of the total revenue. The other 8 locations constitute 69.01% of the remaining revenue.

Implications: High revenues in Chandigarh and Delhi indicate effective market access and strong demand in these areas.

Recommendations:

- Improve distribution networks in low-revenue regions.
- Launch targeted marketing campaigns to increase demand in underperforming areas.



Total Revenue by Farm Size

Large farms generate ₹20.37M (34.68%), medium farms contribute ₹19.18M (32.65%), and small farms add ₹19.19M (32.67%).

Implication: Surprisingly balanced revenue contribution despite different scales

Recommendations:

- Small farms show remarkable efficiency; study their practices for potential scaling.
- Provide training, technology access, and resources to small and medium farms to improve their productivity.

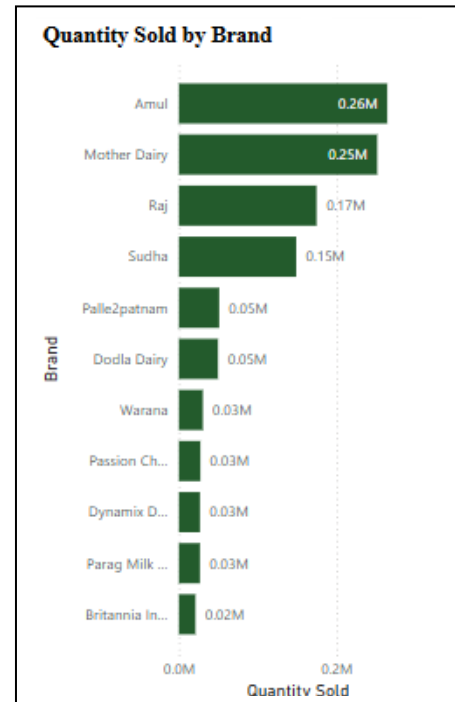
Quantity Sold by Brand

Amul leads with 24.65% (260K units). Mother Dairy is close behind with 23.46% (250K units). Raj and Sudha combined contribute 30.17% (16.29% and 13.88%, respectively). The remaining 7 brands collectively account for 21.72% (170K units).

Implications: Amul and Mother Dairy dominate the market, but smaller brands have limited impact, which could limit overall growth.

Recommendations:

- Invest in marketing for smaller brands to increase visibility and competition.
- Encourage product diversification to create unique selling propositions for less dominant brands.



Brands by Storage Condition

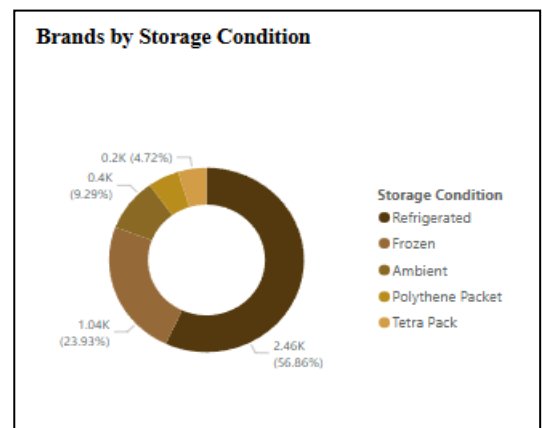
Refrigerated accounts for 56.86% (2,469 units). Frozen contributes 23.93% (1,035 units), while Ambient conditions make up 9.26% (402 units). Polythene packets (5.2%) and Tetra Packs (4.72%) are the least used.

Implications:

- Heavy reliance on refrigerated storage suggests higher energy costs.
- Low use of polythene packets and Tetra Packs may indicate underutilized cost-saving options.

Recommendations:

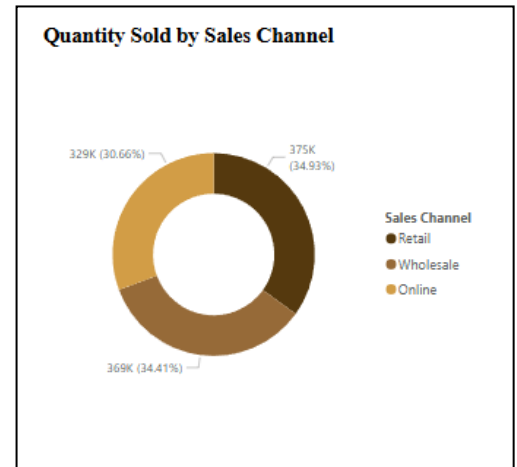
- Explore sustainable, cost-efficient storage solutions like expanding Tetra Pack use.
- Evaluate storage technologies to reduce energy consumption.



Quantity Sold by Sales Channel

- Retail leads with 34.93% (375K units).
- Wholesale accounts for 34.41% (369K units).
- Online sales 30.66% (329K units).

Implications: Wholesale and online channels contribute significantly, but retail remains a key distribution point. This suggests strong market accessibility but may indicate a need to further strengthen retail presence.



Recommendations:

- Leverage digital marketing to further boost online sales.
- Provide incentives to retail outlets to grow this channel and create a balanced distribution.
- Optimize wholesale logistics to maintain cost efficiency.

Location by Quantity Sold

Chandigarh contributes the most at 12.55% (134,625 units). Delhi follows with 12.12% (130,096 units). Lower-performing locations include Madhya Pradesh and Gujarat, each at around 6% (64,531 and 64,015 units, respectively).

Implications: Sales are concentrated in a few locations, suggesting untapped potential in other regions.

Recommendations:

- Expand marketing campaigns in underperforming locations.
- Partner with local distributors to improve penetration in regions like Madhya Pradesh, Gujarat etc.

Product by Quantity Sold

- Curd: 12% of quantity sold (120,415 units).
- Butter: 10.96% (109,578 units).
- Milk: 10.99% (109,852 units).

Implications: Popular products such as curd, butter, and milk drive sales, but lesser-sold items may indicate lower demand or marketing gaps.

Recommendations:

- Focus on promoting lesser-performing products to diversify revenue streams.
- Conduct surveys to understand customer preferences for targeted improvements.

Total Value by Product Name

- Curd: 23.93%
- Butter: 22.27%
- Milk: 21.14%
- Lassi: 21.14%
- Buttermilk: 19.96%

Implications: Curd, Butter, and Milk drive the majority of total value, showcasing customer preference for staples. Lesser-performing products may indicate niche demand.

Recommendations:

- Focus on product innovations for lesser-performing items to attract diverse customer bases.
- Bundle top-performing products with lesser-known ones for promotional offers.
- Conduct surveys to better understand customer preferences and needs.

Product Name	Qty. Sold	Qty. Stocked	Total Value	Total Revenue
Curd	120415	133525	14,090,366.34	6,743,880.73
Butter	109578	115784	12,796,003.03	6,276,041.59
Lassi	112731	113385	12,382,639.94	6,130,168.70
Buttermilk	100402	110789	11,979,054.96	5,767,704.18
Milk	109852	105187	11,735,073.80	6,021,395.90
Paneer	107931	102438	11,518,651.89	5,962,594.91
Yogurt	104744	105490	11,382,504.75	5,595,059.89
Cheese	104065	102112	11,023,898.42	5,547,882.20
Ghee	98513	97858	10,862,445.28	5,385,285.32
Ice Cream	104780	103629	10,552,042.99	5,304,634.44
Total	1073011	1090197	118,322,681.40	58,734,647.86

Products by Storage Condition

- Refrigerated: 56.86%
- Frozen: 23.93%
- Ambient: 9.29%
- Polythene Packets: 5.2%
- Tetra Packs: 4.72%

Implications: Refrigerated storage dominates, highlighting a preference for preserving product freshness. However, a significant gap in the use of sustainable packaging options (Tetra Packs and Polythene Packets) may lead to higher environmental and operational costs.

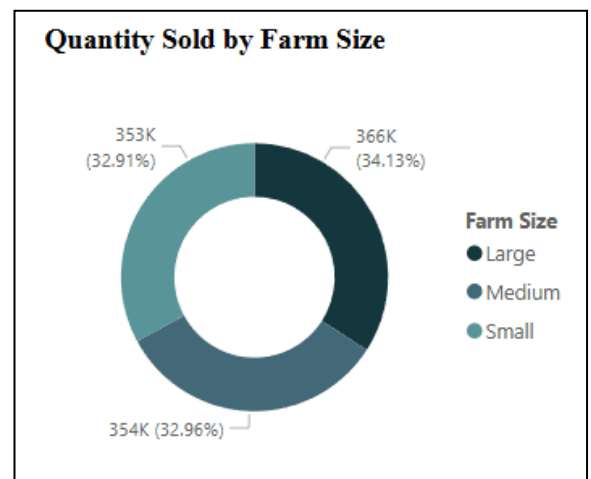
Recommendations:

- Invest in sustainable packaging solutions like Tetra Packs.
- Optimize refrigerated storage systems to reduce energy consumption and costs.
- Explore ambient storage improvements for products with longer shelf lives.

Quantity Sold by Farm Size

- Large Farms: 34.13%
- Medium Farms: 32.96%
- Small Farms: 32.91%

Implications: Large farms have a slight edge due to economies of scale, while small and medium farms account for nearly two-thirds of total production. This indicates growth potential in small and medium-scale farming.



Recommendations:

- Provide advanced training and resources to small farms to increase their production efficiency.
- Offer financial incentives or subsidies to medium farms for expansion.

Product by Shelf-Life Distribution

- 21-30 days: 28.95%
- 0-10 days: 25.87%
- 11-20 days: 20.99%
- 31-40 days: 7.58%
- Others (41-100 days): Remaining 16.61%

Implications: Products with shorter shelf lives dominate, signaling a focus on fresher goods. Longer shelf-life products are underutilized, possibly due to lower demand or marketing gaps.

Recommendations:

- Prioritize marketing for longer shelf-life products to diversify sales.
- Ensure efficient distribution to minimize waste for shorter shelf-life goods.
- Develop partnerships with storage providers to maintain product quality over time.

Shelf Life	%GT of Products
21 - 30	28.95%
0 - 10	25.87%
11 - 20	20.99%
31 - 40	7.58%
61 - 70	2.66%
71 - 80	2.34%
81 - 90	2.24%
51 - 60	1.69%
41 - 50	1.55%
141 - 150	1.16%
131 - 140	1.11%
91 - 100	1.09%
111 - 120	1.06%
101 - 110	0.86%
121 - 130	0.86%
Total	100.00%

Conclusion

The dairy operation demonstrates a well-distributed business model across farm sizes, product categories, and sales channels. The balanced revenue generation across farm sizes suggests effective operational management regardless of scale. Northern regions, particularly Chandigarh and Delhi, represent the most productive and profitable locations for the business.

The dominance of traditional products like curd, butter, and lassi in both revenue and value indicates strong consumer preference for staple dairy items. The nearly equal distribution across sales channels shows successful market penetration across retail, wholesale, and online segments, suggesting an adaptive distribution strategy.

For future growth, the operation might consider expanding refrigerated storage capacity given its prevalence, and potentially developing products with longer shelf

life to complement the current portfolio that is heavily concentrated in short shelf-life categories.